YASHWANTH REDDY MALI

Software Engineer, Backend

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SUMMARY

Backend engineer who has shipped production REST APIs, authentication, and CI/CD across Node.js and C#/.NET, and builds distributed systems in Go. Found and privately disclosed a fail-open authorization vulnerability in OpenFGA, a CNCF-incubating engine used by Grafana and Docker, and wrote the fix that shipped in v1.19.0.

EXPERIENCE

Credible Data - Software Engineer *Boulder, CO | Aug 2025 - May 2026 (part-time)*

- Built REST APIs in Node.js and Express on a five-person Agile team, routing a single query across PostgreSQL, Neo4j, and BigQuery; deployed on GCP Cloud Run through GitHub Actions CI/CD with workload identity federation.
- Implemented OAuth2/OIDC authentication and three-tier RBAC with Auth0, with scheduled admin-token refresh out of Google Secret Manager and 401 retry handling against the upstream provisioning API, so a provisioning failure routes to a retry endpoint rather than the sign-in path.
- Designed event-driven sync from PostgreSQL to Neo4j with the transactional outbox pattern, committing the outbox row in the same transaction as the business write so an unreachable graph never fails a request.

IntelleWings - Software Engineer Intern *Chandigarh, India (Remote) | Feb 2023 - Jun 2024*

- Migrated a nightly anti-money-laundering screening batch to Apache Spark, processing 10 million records in 60 minutes over an ETL pipeline loading CSV extracts into MySQL.
- Converted the platform from single-tenant to multi-tenant, cutting infrastructure cost by isolating every customer in one deployment instead of a stack each; built the ingestion parser aggregating 230+ external data sources and migrated the C#/.NET codebase from .NET Core 5.0 to 8.0.

OPEN SOURCE

OpenFGA - Upstream Contributor | CNCF-incubating authorization engine, built by Okta *Go*

- Disclosed and fixed a fail-open authorization vulnerability in streaming evaluation, shipped in v1.19.0 (GitHub Security Advisory GHSA-5278-rrxc-mgf7): an exclusion clause erroring mid-evaluation let StreamedListObjects emit objects whose access check never completed.
- Fixed a Check API correctness bug in v1.18.2 (PR #3225) where the query flattening pass tracked recursion as a single boolean, with regression tests covering both recursive relations.

PROJECTS

Interlock - Runtime security layer for AI agent infrastructure *Go, eBPF, Kubernetes, SQLite*

- Built a Go proxy and Linux eBPF/LSM sensor tracking tainted data across 13 encodings, confirming exfiltration by matching the bytes that actually left rather than pattern heuristics; 336 Go tests behind CI race and benchmark gates.
- Evaluated the enforcement layer against 15 attack reconstructions spanning 7 CVE families: detected 12, zero false positives across 37 benign scenarios, and three documented detection gaps. Ships as a Kubernetes DaemonSet with Prometheus metrics.

Dispatch - Event delivery platform with at-least-once guarantees *Go, Kafka, PostgreSQL, Redis, Terraform, AWS*

- Built an event-driven microservice with at-least-once delivery, idempotency keys, a Postgres dead-letter queue, and a circuit breaker; every retry rides a single Kafka topic keyed by retry_after, so adding a retry tier is a config change, not new infrastructure.
- Validated crash recovery by SIGKILLing the consumer under load, then reconciling all 16,001 published events against delivery-attempt and dead-letter records: zero events left without an attempt. Provisioned on AWS with Terraform (EKS with IRSA, MSK, RDS, S3), deployed with Helm, instrumented with Prometheus.

Zenith - Distributed authorization database with consensus replication *Go, gRPC, Raft, MVCC*

- Built a Go authorization database with Raft replication from the paper (pre-vote, joint-consensus membership, learners, ReadIndex, log compaction, InstallSnapshot) across 3 to 5 nodes; checked KV operation histories for linearizability with Porcupine across 2,000 deterministic simulation seeds.

Ratchet - Incremental ELT sync engine and connector SDK *Java 21, JUnit, Apache Arrow, Parquet*

- Built an incremental ELT engine in Java 21 whose sync cursor advances only after the data is durable, so every crash window replays work instead of dropping rows. Verified at 500,000 rows through 50 seeded SIGKILLs, checksums matching.

Also: LineageGraph (Python, DuckDB), natural-language data lineage over a local LLM with a golden-dataset eval harness.

TECHNICAL SKILLS

Languages: Java, Python, Go, TypeScript, JavaScript, C# and .NET, SQL, C
Backend: REST APIs, microservices, Node.js, Express, gRPC, API design, OAuth2/OIDC, JWT, RBAC, event-driven architecture, message queues
Databases and Data: PostgreSQL, MySQL, Redis, Kafka, Neo4j, BigQuery, SQLite, DuckDB, Apache Spark, ETL and ELT
Cloud and DevOps: AWS (EKS, MSK, RDS, S3, IRSA), GCP (Cloud Run, BigQuery), Docker, Kubernetes, Helm, Terraform, CI/CD with GitHub Actions, Prometheus, Linux, eBPF
Testing: Go testing and fuzzing, JUnit, linearizability checking (Porcupine), deterministic simulation, fault injection

EDUCATION

University of Colorado Boulder - M.S. Computer Science, GPA 3.9/4.0 *Aug 2024 - May 2026*

Osmania University - B.E. Artificial Intelligence and Data Science, GPA 8.7/10.0 *Aug 2020 - May 2024*